

Member ID: _____

Time: _____

Rank: _____



NETWORK ADMINISTRATION USING CISCO (315)

STATE 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each) _____ (100 points)

TOTAL POINTS _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

MULTIPLE CHOICE

Identify the letter of the choice that *best* completes the statement or answers the question.

1. What is a function of a router?
 - a. It connects multiple IP networks.
 - b. It controls the flow of data via the use of Layer 2 addresses.
 - c. It provides segmentation at Layer 2.
 - d. It builds a routing table based on ARP requests.
2. Which one of the following IP address settings is needed to be able to access the internet but is not needed for local communication?
 - a. Gateway
 - b. IP Address
 - c. Subnet Mask
 - d. DNS Server
3. How many bits does an IPV6 address have?
 - a. 32
 - b. 48
 - c. 128
 - d. 16
4. Which of the following protocols is considered a layer-4 protocol?
 - a. Ethernet
 - b. IP
 - c. TCP
 - d. HTTP
5. SSH allows a user to connect securely to a remote device and ensures all communication is encrypted
 - a. False
 - b. True
6. FTP and TFTP are both allow safe file transfer functionalities
 - a. True
 - b. False
7. What device divides broadcast domains?
 - a. Router
 - b. Switch
 - c. Bridge
 - d. Repeater

8. Which of the following port modes describes a port that allows several VLANs to go through?
 - a. Access Port
 - b. Trunk Port
 - c. Secure Port
 - d. General Port
9. Are systems on different VLANs able to communicate between each other without a Router?
 - a. True
 - b. False
10. Which device determines the path that messages should take through internetworks?
 - a. Router
 - b. Firewall
 - c. Web server
 - d. Switch
11. Which of the following port modes describes a port that allows a single VLAN untagged to go through?
 - a. Access Port
 - b. Trunk Port
 - c. Secure Port
 - d. General Port
12. Which field in the IPv4 header is used to so that a does not travel over the network endlessly?
 - a. Time-to-Live
 - b. Sequence Number
 - c. Acknowledgment Number
 - d. Differentiated Service
13. What is one advantage that IPv6 offers over IPv4?
 - a. Smaller-sized header
 - b. Little requirement for processing checksums
 - c. Smaller-sized source and destination IP addresses
 - d. Efficient packet handling

14. What is the next step in the router boot sequence after the IOS loads from flash?
 - a. Perform the POST routine.
 - b. Locate and load the startup-config file from NVRAM.
 - c. Load the bootstrap program from ROM.
 - d. Load the running-config file from RAM.
15. What command is used to display the IP addresses assigned on the switch/router?
 - a. show ip interface brief.
 - b. show ip.
 - c. show addresses.
 - d. show interface state.
16. What does the IP address 172.17.4.250/24 represent?
 - a. Network address
 - b. Multicast address
 - c. Host address
 - d. Broadcast address
17. What type of address is automatically assigned to an interface when IPv6 is enabled on that interface?
 - a. Global unicast
 - b. Link-local
 - c. Loopback
 - d. Unique local
18. What is a reason a network administrator might want to create subnets?
 - a. Simplifies network design.
 - b. Allow more networks to be made.
 - c. Reduction in number of routers needed.
 - d. Reduction in number of switches needed.
19. What happens if two or more switches are connected together?
 - a. The number of broadcast domains is increased.
 - b. The size of the broadcast domain is increased.
 - c. The number of collision domains is reduced.
 - d. The size of the collision domain is increased.
20. How many assignable host addresses are available on the network 172.16.128.0 with a subnet mask of 255.255.252.0?
 - a. 510
 - b. 512
 - c. 1022
 - d. 1024

21. Which feature allows different communication streams to occur at the same time without a single stream using the entire bandwidth?
- Window size
 - Multiplexing
 - Port numbers
 - Acknowledgments
22. What type of information is included in the transport header?
- Destination and source logical addresses
 - Destination and source physical addresses
 - Destination and source port numbers
 - Encoded application data
23. Which task can be performed by a local DNS server?
- Providing IP addresses to local hosts.
 - Allowing data transfer between two network devices.
 - Mapping name-to-IP addresses for internal hosts.
 - Retrieving email messages.
24. Which application layer protocol is used to provide file-sharing and print services to Microsoft applications?
- HTTP
 - SMTP
 - DHCP
 - SMB
25. Which type of network threat floods the network with requests so that users cannot access resources?
- DoS attacks
 - Access attacks
 - Reconnaissance attacks
 - Trust exploitation
26. If the router is using an information base and an adjacency table, it means it's using which packet-forwarding method?
- Fast switching
 - Cisco Express Forwarding
 - Process switching
 - Flow process
27. What route would have the lowest administrative distance?
- Directly connected network.
 - Static route.
 - Route received through the EIGRP routing protocol.
 - Route received through the OSPF routing protocol.

28. What is a commonly seen type of static route?
- a. Default static route.
 - b. Built-in static route by IOS.
 - c. Static route shared between two neighboring routers.
 - d. Static route converted from a route that is learned through a dynamic routing protocol.
29. If an outgoing interface that has a static route associated to it goes down, what happens to the route?
- a. The static route is removed from the routing table.
 - b. The router polls neighbors for a replacement route.
 - c. The static route remains in the table because it was defined as static.
 - d. The router automatically redirects the static route to use another interface.
30. Which dynamic routing protocol was created to interconnect different ISPs?
- a. BGP
 - b. EIGRP
 - c. OSPF
 - d. RIP
31. Which routing protocol is used on smaller network implementations because it does not work properly on bigger networks?
- a. OSPF
 - b. RIP
 - c. EIGRP
 - d. IS-IS
32. Why is the passive-interface command used?
- a. Allows a router to send routing updates on an interface but not receive updates via that interface.
 - b. Allows an interface to remain up without receiving keepalives.
 - c. Allows interfaces to share IP addresses.
 - d. Allows a router to receive routing updates on an interface but not send updates via that interface.
33. Which level of the OSI 7-layer model is the Transport Layer?
- a. 7th Level
 - b. 5th Level
 - c. 4th Level
 - d. 3rd Level

34. Which IP Address below is a valid private Class-A IP Address?
- a. 10.255.75.23
 - b. 10.255.256.1
 - c. 172.16.13.4
 - d. 192.168.100.83
35. Which IP Address below is a valid Class-B IP Address?
- a. 172.16.13.4
 - b. 172.17.10.1
 - c. 11.0.52.1
 - d. 192.168.100.83
36. Which IP Address below is a valid Class-C IP Address?
- a. 192.168.100.83
 - b. 192.168.0.257
 - c. 11.0.52.1
 - d. 172.16.13.4
37. Which protocol uses UDP?
- a. DNS
 - b. FTP
 - c. HTTP
 - d. SSH
38. In what type of attack may these tools (nslookup and fping) be involved?
- a. Access attack
 - b. Reconnaissance attack
 - c. Denial of service attack
 - d. Worm attack
39. What is a must for an administrator to be able to reset a router's password?
- a. TFTP server
 - b. Crossover cable
 - c. Access to another router
 - d. Physical access to the router
40. What is it used to locate and load the IOS image?
- a. BOOT environment variable
 - b. IOS image file
 - c. POST
 - d. startup-config

41. What information is contained in the 12-bit extended system ID of a BPDU?
- MAC address
 - VLAN ID
 - IP address
 - Port ID
42. If there is no other port with a lower cost to the root bridge, which port role is used on the switch port?
- Designated port
 - Root port
 - Alternate
 - Disabled port
43. If no bridge priority is configured in PVST, based on what is the root bridge selected?
- Lowest MAC address
 - Lowest IP address
 - Highest IP address
 - Highest MAC address
44. Which switching method ensures is used to prevent forwarding frames with errors?
- Cut-through
 - FCS
 - Fragment free
 - Store-and-forward
45. Which cable is used to connect to a switch to manage it?
- Roll-over Cable
 - Straight Cable
 - Serial Cable
 - Crossover Cable
46. What happens to switch ports if you delete the VLAN that was assigned to it?
- The ports are disabled.
 - The ports are placed in trunk mode.
 - The ports are assigned to VLAN1, the default VLAN.
 - The ports stop communicating with the attached devices.
47. Which mode is being used when the prompt shows “Switch(config)#”?
- Interface Configuration
 - User Exec
 - Privilege Exec
 - Global Configuration

48. What does the reload command do on Privilege EXEC mode?
- a. Reboots the device without saving the configuration.
 - b. Reboots the device while saving the configuration.
 - c. Reloads the saved configuration only.
 - d. Rearms the security enabled on the ports of the device.
49. A host receives a frame and discards it after determining it is corrupt. Which OSI layer does this?
- a. Application
 - b. Physical
 - c. Data-Link
 - d. Network
50. Which port type can use the spanning-tree portfast command without needing any other configuration?
- a. Access ports
 - b. Layer 3 main interfaces
 - c. Layer 3 subinterfaces
 - d. Trunk ports